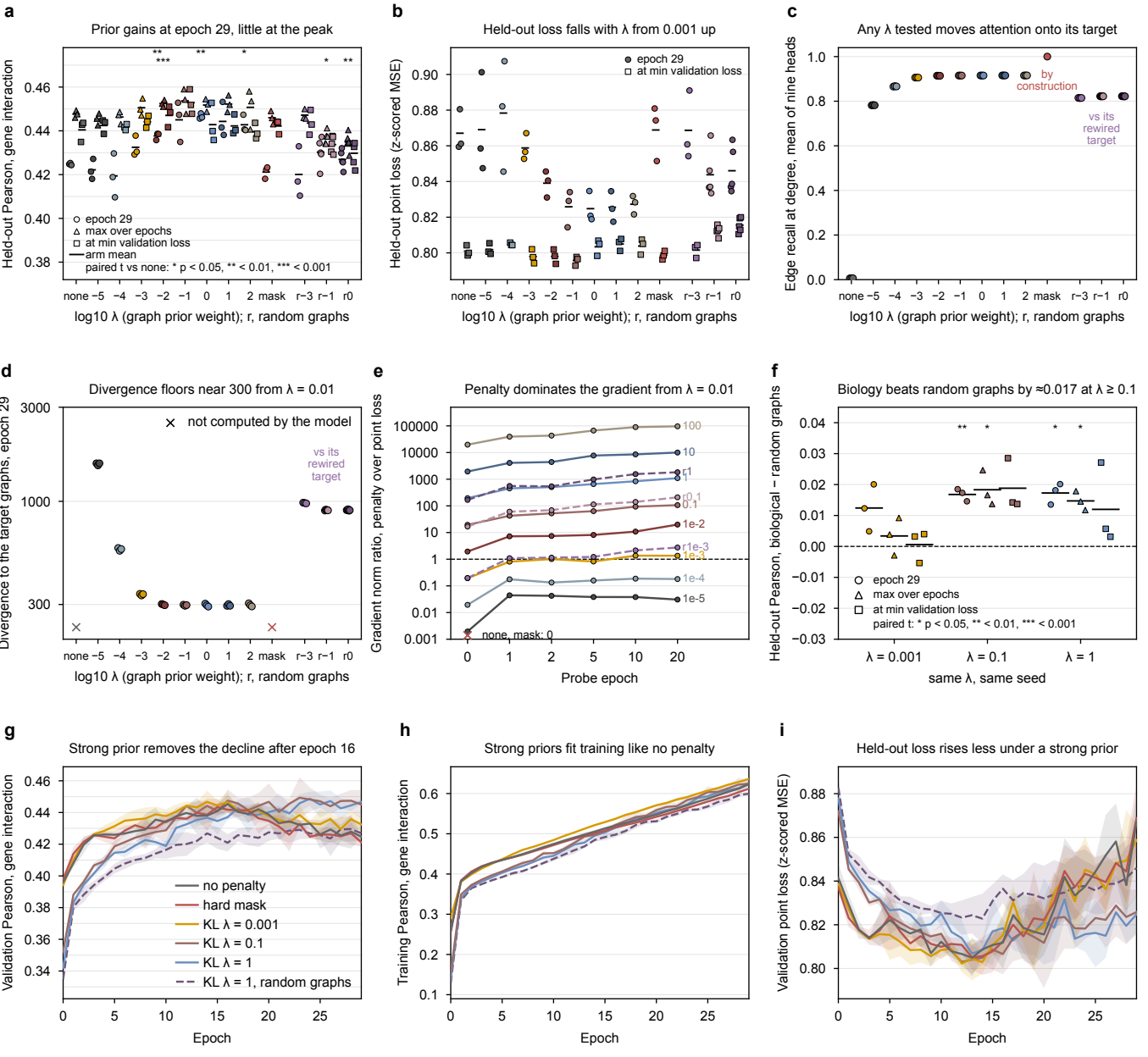

a
Prior gains at epoch 29, little at the peak
Held-out Pearson, gene interaction
epoch 29
max over epochs
at min validation loss
arm mean
paired t vs none: * p < 0.05, ** < 0.01, *** < 0.001
log10 λ (graph prior weight); r, random graphs
b
Held-out loss falls with λ from 0.001 up
Held-out point loss (z-scored MSE)
epoch 29
at min validation loss
log10 λ (graph prior weight); r, random graphs
c
Any λ tested moves attention onto its target
Edge recall at degree, mean of nine heads
by construction
vs its rewired target
log10 λ (graph prior weight); r, random graphs
d
Divergence floors near 300 from λ = 0.01
Divergence to the target graphs, epoch 29
not computed by the model
vs its rewired target
log10 λ (graph prior weight); r, random graphs
e
Penalty dominates the gradient from λ = 0.01
Gradient norm ratio, penalty over point loss
100
10
1
0.1
1e-2
1e-3
1e-4
1e-5
none, mask: 0
Probe epoch
f
Biology beats random graphs by ≈0.017 at λ ≥ 0.1
Held-out Pearson, biological − random graphs
epoch 29
max over epochs
at min validation loss
paired t: * p < 0.05, ** < 0.01, *** < 0.001
λ = 0.001
λ = 0.1
λ = 1
same λ, same seed
g
Strong prior removes the decline after epoch 16
Validation Pearson, gene interaction
no penalty
hard mask
KL λ = 0.001
KL λ = 0.1
KL λ = 1
KL λ = 1, random graphs
Epoch
h
Strong priors fit training like no penalty
Training Pearson, gene interaction
Epoch
i
Held-out loss rises less under a strong prior
Validation point loss (z-scored MSE)
Epoch